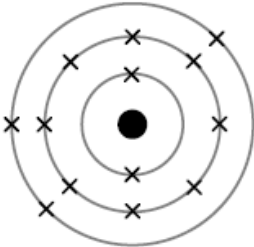


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			 accept electrons shown as dots accept diagram with nucleus missing		1		1		
	(b)	(i)		B and D (1) must be correct to access second mark award (1) for either of following they have same number of protons but different number of neutrons they have same atomic number but different mass number ignore reference to electrons	1	1		2		
		(ii)		A and F (1) must be correct to access second mark award (1) for any of following they have different numbers of protons and electrons A has more electrons than protons and F has more protons than electrons neutral answer - they are A⁻ and F⁺ do not award the mark if there is any suggestion that the number of neutrons is relevant	1	1		2		
				Question 5 total	2	3	0	5	0	0